



Extraction of entity relationships serving the field of agriculture food safety regulation

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Received: 4 September 2023 / Accepted: 26 July 2024 / Published online: 21 August 2024
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Abstract

Agriculture food (agri-food) safety is closely related to all aspects of people's lives. In recent years, with the emergence of deep learning technology based on big data, the extraction of information relations in the field of agri-food safety supervision has become a research hotspot. However, most of the current work only expands the relationship recognition based on the traditional named entity recognition task, which makes it difficult to establish a true 'connection' between entities and relationships. The pipelined and federated extraction architectures that have emerged in this area are problematic in practice. In addition, the contextual information of the text corpus in the agri-food safety regulatory domain has not been fully utilized. To address the above issues, this paper proposes a semi-joint entity relationship extraction model (EB-SJRE) based on contextual entity boundary features. Firstly, a Token pair subject-object correspondence matrix label is designed to intuitively model the subject-object boundary, which is more friendly to complex entities in the field of agri-food safety regulation. Secondly, the dynamic fine-tuning of Bert makes the text embedding more relevant to the textual context of the agri-food safety regulation domain. Finally, we introduce an attention mechanism in the Token pair tagging framework to capture deep semantic subject-object boundary association information, which cleverly solves the problem of bias exposure due to the pipeline structure and the dimensional explosion due to the joint extraction structure. The experimental results show that our model achieves the best F1-score of 88.71% on agri-food safety regulation domain data and F1-scores of 92.36%, 92.80%, 88.91%, and 92.21% on NYT, NYT-star, WebNLG, and WebNLG-star, respectively. This indicates that EB-SJRE has excellent generalization ability in both the agri-food safety regulatory and public sectors.

Keywords Relation extraction · Agriculture food safety regulation · Attention mechanisms · RCNN · Bert

1 Introduction

With the emergence of deep learning technologies based on big data, the field of agricultural food safety supervision [1] has seen new developments. Information extraction in this

field has become an indispensable research focus, as it is the basis for downstream tasks such as knowledge graph construction [2], data network mining [3], and safety traceability chains [4]. However, current research related to intelligent regulation is more focused on the application of computer vision in edible agricultural products [5–7]. Alternatively, feedback from users who purchased the produce can be utilized for product quality analysis [8, 9]. There is little literature that starts from specific agricultural food safety standards and regulations, knowledge, and regulatory information [10]. This information is scattered and distributed, contains a large amount of regulatory knowledge, and is essential for guiding agricultural food safety risk analysis and aiding research and judgment [11]. Therefore, it is necessary to propose a set of underlying information extraction schemes applicable to the agricultural food safety regulatory field to correlate siloed information.

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In recent years, deep learning-based entity relationship extraction schemes have achieved significant breakthroughs in various fields, including medicine and health [12], geology [13], finance [14], biomedicine [15], and agriculture [16]. These schemes do not rely excessively on annotated information and manual feature engineering, making them more versatile and efficient. Current methods for entity relationship extraction are mainly from two perspectives: feature embeddings [17, 18] and feature encoders [19, 20]. In addition to better explaining the extraction process of the model, entity relationship extraction can be further categorized into pipeline approach [21] and joint approach [22, 23]. However, in the field of agricultural food safety regulation, entity relationship extraction faces challenges due to inadequate handling of mixed-language entities and domain-specific terms, coupled with significant biases introduced by traditional methods. So, there is an urgent need for specified algorithms customized to suit the unique data characteristics of the domain.

This paper introduces a semi-joint relationship extraction model based on contextual entity boundary features (EB-SJRE) to overcome the problems in previous entity relationship extraction models. Firstly, EB-SJRE uses a Token pair subject-object correspondence matrix label to correlate the boundaries of the first and last entities of the corpus. This eliminates the need for the model to focus on the interior of entities and only requires it to focus on the first and last entity boundaries, which is more relevant for complex entities in the agri-food safety regulation domain. Secondly, to ensure the effectiveness of entity-to-association, we propose a TPA boundary attention module incorporating potential relationships to model the semantic associations between entity boundaries. Finally, we combine the above solutions and propose a semi-joint entity relationship extraction model, which first identifies the types of relationships contained in the text and then jointly identifies the information of entity pairs under the corresponding relationships by incorporating golden relationship features. This not only reduces the bias exposure brought about by the traditional pipeline model but also solves the problems of dimensional expansion and label sparsity brought about by modeling relational entities in the joint extraction model. In summary, the main contributions of this paper are as follows:

1. We propose a Token pair first and last entity cross-reference matrix labeling approach for domain datasets. This approach discards the traditional BIO sequence approach and focuses model attention on capturing the contextual boundaries of the first and last entities, which is more friendly to complex entities in the agri-food safety regulation domain.
2. We propose a semi-joint extraction model of entity relations based on fine-tuned Bert, which can effectively

learn domain semantic information embedded in the first stage with RCNN [24] support and effectively model the semantic association of entity boundaries in the second stage with the TPA boundary attention module incorporating potential relations. This can effectively reduce the problem of bias exposure caused by the traditional pipeline structure and also solve the problems of dimensional expansion and label sparsity caused by the modeling of relational entities in the joint extraction model.

3. We have conducted a number of comparative experiments with typical joint extraction models published in recent years on agri-food safety regulatory domain data as well as public domain data. The results demonstrate that EB-SJRE is not only very effective in the agri-food safety regulatory domain but also has very strong generalization capabilities.

2 Related work

Entity relationship extraction is used to identify and link the relevant entities in the text by understanding the context and the relationship between them to help analyze and organize data. At this stage, entity-relationship extraction algorithms have been explored by research scholars in different fields [14, 15]. According to the different focuses of the research, they are further divided into two directions: 'data representation and encoding' and 'data extraction strategy'. The former focuses on data preprocessing and representation methods, e.g., how to effectively encode entities and relationships through embedding techniques, while the latter focuses on information or data extraction methods, exploring how to accurately identify and extract relevant entities and their relationships from complex datasets. We have organized a table of related work by referring to the methodology adopted by Akkem et al. [25] in their article, as shown in Table 1. It lists the main algorithms of the following mentioned entity-relationship extraction studies and gives their authors, year of publication, and journal of publication, as well as possible shortcomings in their application to the field of agri-food safety.

2.1 Data representation and encoding

Entity relationship extraction methods in these fields are generally studied from two perspectives: (1) feature embeddings: current embedding methods include random initialization and pre-training based Word2Vec [17] and Bert [18], which are closely related to the complexity and size of the data [32]. Domain improvement methods generally fuse character- or word-level embedding features, and there are also improvements from the perspective of multivariate information fusion, such as the introduction of syntactic

Table 1 Related work analysis table

Research Focus	Main Methods	Author	Description	Source	Disadvantages
Embedding features	Word2Vec[26]	Jang et al. (2019)[17]	Relevance classification of news articles and tweets using word2vec and convolutional neural networks (CNN).	PloS one	They are closely related to the complexity and size of the data
	Bert	Devlin et al. (2018)[18]	BERT pre-trains deep bidirectional representations from unlabeled text, considering both left and right context in all layers.	arXiv	
	Syntax-enhanced model	Liu et al.(2022)[27]	A syntactic enhancement model using category keywords.	Journal of Biomedical Informatics	
	CG-ANER	Guo et al. (2022)[16]	Agricultural NER models based on contextual embeddings and glyph features.	Computers and Electronics in Agriculture	
Feature encoders	Bi-LSTM	Munkhdalai et al.(2018)[19]	Evaluating natural language processing and machine learning approaches for the use of medical entities and relationships in drug monitoring.	JMIR public health and surveillance	Unsuitable for dealing with complex entity and multidimensional relationship feature cases.
	PCNN	Wen et al. (2020)[20]	A Gated Piecewise CNN with Entity-Aware Enhancement for distantly supervised relationship extraction.	Information Processing & Management	
	GCN	Hong et al. (2020)[28]	An end-to-end neural model based on Graph Convolutional Networks for jointly extracting entities and relationships between entities.	IEEE Access	
Survey	\	Liu et al. (2020)[29]	A survey on the task of neural relation extraction.	Science China Technological Sciences	\
Pipeline structures	FHER	Zuo et al. (2022)[21]	An entity relation extraction method FHER for the few-shot learning in the food health and safety domain.	Computational Intelligence and Neuroscience	May lead to subsequent error amplification and serious bias exposure.
Joint structures	CasRel	Wei et al. (2019)[22]	A novel cascade binary tagging framework derived from a principled problem formulation.	arXiv	The model seems logically unsound in the "relationship-object" perspective.
	BIRTE	Ren et al. (2022) [23]	A bidirectional extraction framework based method that extracts triples based on the entity pairs extracted from two complementary directions.	In Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining	
	OneRel	Shang et al. (2022) [30]	The model treats joint extraction as a fine-grained triple classification problem.	In Proceedings of the AAAI Conference on Artificial Intelligence	When the number of relationships is high, model size proliferation and label sparsity may occur.
	Tp-Linker	Wang et al. (2020)[31]	A one-stage joint extraction model for discovering overlapping relations sharing one or both entities.	arXiv	A significant decrease in accuracy compared to the single-step decoding

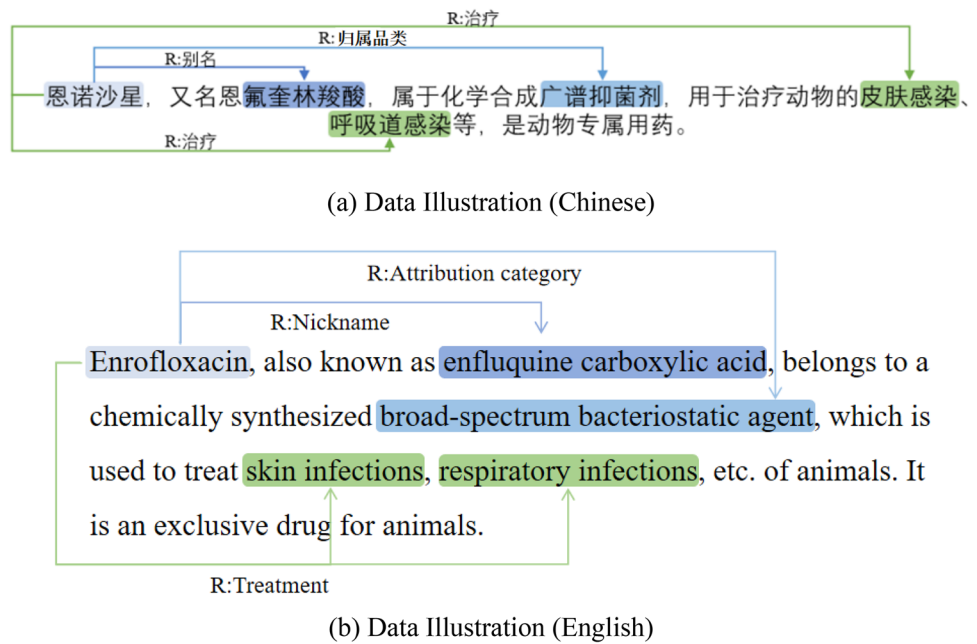
structure features of the corpus [27] or word form features [16]. These methods have been proven to be effective in specific application scenarios. (2) Feature encoders: feature encoders, such as the Bi-LSTM model [19] represented by the RNN series, can effectively extract contextual sequence information to obtain more detailed features for entity relationship extraction tasks. However, its operation speed is relatively slow. The PCNN [20] model represented by CNN can effectively capture local feature information of the corpus with moderate speed. At the graph neural network level, the GCN [28] model represented by GNN can better capture the global word co-occurrence information and has a faster operation speed.

Although the above schemes can also be used for information extraction in the agri-food safety regulation domain, they may not be suitable for dealing with the complex entity and multidimensional relational feature scenarios, as shown in Fig. 1. This is because they mainly focus on character, word, location, or statement-level feature engineering, which may not fully capture the unique characteristics of the agri-food safety regulation domain. Additionally, these methods may lack interpretability, making it difficult to understand how they arrived at their results.

2.2 Data extraction strategy

In order to achieve better interpretability, research on entity relationship extraction has shifted toward the study of model architectures and labels. These studies can be divided into pipeline structures and joint extraction structures [29]. The pipeline structure, which generally extracts entities before relations, has been widely used in the past. For example, Zuo et al. first sequentially identify entities and then simultaneously incorporate their features to identify corresponding relations, successfully applying this approach in the field of agri-food [21]. However, this scheme has a serious problem in that two entities and their corresponding relations are identified sequentially, and features are passed to the next step. If the previous recognition is incorrect, it can lead to subsequent error amplification and serious bias exposure. The joint extraction structure, which models the relevance of relations and entities, usually identifies the subject in the text first and then incorporates its features to model the relations and objects spatially. For example, the CasRel model proposed by Wei et al. does this [22]. Ren et al. constrain the entities on their basis [23], and Zuo et al. successfully apply this relation-object modeling idea to the field of food

Fig. 1 Sample text of entity-relationship in the field of agri-food safety regulation. **(a)** Data illustration (Chinese), **(b)** data illustration (English)



health [21]. Although this scheme is good for improving the shortcomings of the pipeline structure and reducing multiple bias exposure, its modeling from the "relation-object" perspective seems to be logically unsound, which is one of the reasons for its moderate effect. The OneRel [30] model proposed by Shang et al. directly extracts triadic information in the text in a one-step manner by spatial modeling "[relation-subject-object]." However, its designed labels force the corpus length to increase by a factor of 1 because it cannot handle single-word entities, so it is necessary to construct a high-dimensional matrix of "[number of relations $\times 2 \times$ text length $\times 2 \times$ text length]" When the number of relations is high, this approach brings problems such as the proliferation of model size and sparse labels. The Tp-Linker [31] model proposed by Wang et al. also models the "[relation-subject-object]," but it compresses the high-dimensional matrix of OneRel, making training inference faster. However, due to the multi-step decoding approach leads to a significant decrease in accuracy compared to the single-step decoding of OneRel.

2.3 Document analysis

Furthermore, we deeply study the application of heuristic algorithms in our research, such as DETDO [33] and the firefly algorithm [34]. We note that these algorithms provide innovative methods for engineering optimization and control systems, but their direct application or adaptability in our specific tasks is limited. Overall, research on entity relationship extraction is becoming more sophisticated. However, the specific area of agri-food safety regulation is still in its infancy, and there are many issues that need to be addressed.

1. In previous entity relationship extraction models, entity tags in the named entity identification stage usually use the traditional BIO annotation form or its variants. However, this sequential tagging approach often fails to achieve the expected results when dealing with a large number of entities consisting of Chinese characters, letters, and numbers mixed in the text, which is unique to the agri-food safety regulatory domain.
2. Some pre-trained word embedding models based on Word2Vec or Bert may have reduced generalization ability without sufficient domain pre-trained corpus to support them. For example, the detection indicator "胡萝卜素" (carotene) can be captured by a generic pre-trained model as "胡萝卜" (carrot), but not as the complete concept of "胡萝卜素"(carotene), which can cause misclassification.
3. In the complex hierarchy of textual relationships in the field of agri-food safety regulation, the traditional pipeline structure can cause severe bias exposure when performing segmental information extraction. This, in turn, leads to severe degradation in extraction quality. On the other hand, the joint extraction scheme considers the connections between entities or between entities and relationships. However, it usually raises the spatial dimension of the model by one level, which can cause problems such as excessive volume, slowing down of training inference, and sparse labels.

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relationship. For example, "恩" serves as the start marker of "恩诺沙星" (Enrofloxacin), and "酸" is the end marker of "恩氟奎林羧酸" (Enrofloxacin Carboxylic Acid). Therefore, the combination ("恩", "别名", "酸") is designated as the marker "M." (3) T, an associated subscript that denotes the termination marker of a subject and an object, conditioned on a specific relationship. For instance, "星" is the end marker of "恩诺沙星" (Enrofloxacin) and "染" is the end marker of "皮肤感染"(skin infections). Therefore, the combination ("星", "治疗", "染") is designated as the marker "T." (4) -, all other positions in the label matrix are set to "-" except for the three relationships mentioned above.

3.2 Model architecture

The architecture of the EB-SJRE model proposed in this paper is illustrated in Fig. 4. It comprises three modules, namely, embedding, potential relationship prediction, and entity pair prediction.

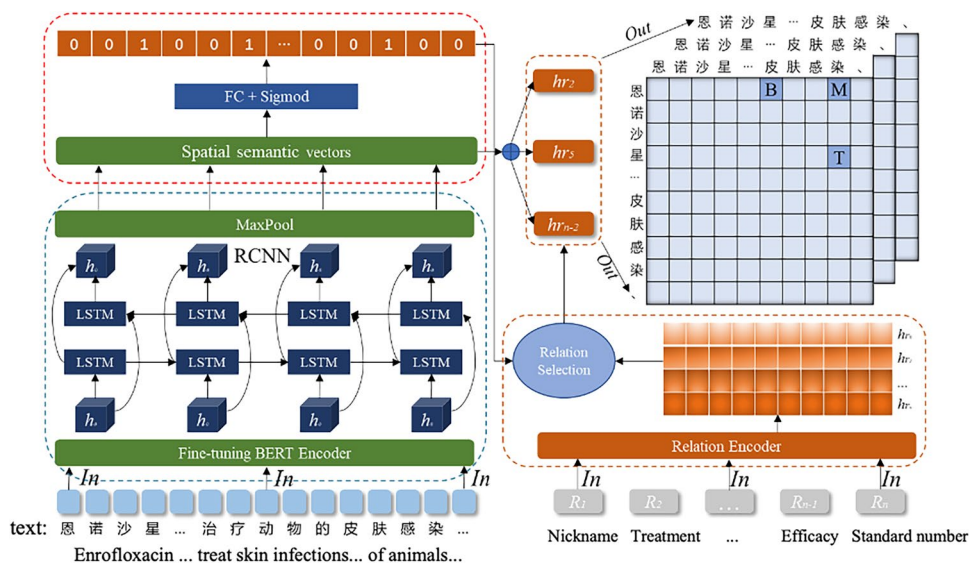
3.2.1 Embedding layer

The embedding layer is utilized to convert each character into a high-dimensional embedding representation. Specifically, for a given corpus $S = (c_1, c_2, \dots, c_{n-1}, c_n)$ of length n , the embedding vector $E = (h_0^d, h_1^d, \dots, h_{n-1}^d, h_n^d)$ is obtained after passing through the embedding layer, where the embedding vector h_n^d corresponds to each character embedding in the S statement, and d is the embedding dimension. While Word2Vec has been widely used as a character embedding in previous related work, it lacks a strong contextual link and cannot effectively represent multiple-meaning words such as "Apple" in fruit and

"Apple" in Apple Inc. To address this issue, we employ a more advanced Chinese pre-trained language model, namely, Bert, as the underlying contextual embedding, which can effectively exploit contextual information. However, the existing Chinese Bert is trained based on a public corpus, and there are significant differences in the distribution of text sequences between the agri-food safety regulatory domain and the public domain. Therefore, fine-tuning Bert for domain-specific texts is necessary and offers the following benefits: (1) it helps Bert to learn contextual semantic information that is relevant to agri-food safety regulation and improves the quality of the embedded representation; (2) it assists downstream tasks in distinguishing between related entities that are similar in name but far apart in meaning, such as "胡萝卜" (edible agricultural product) and "胡萝卜素"(nutrient).

During the process of selecting the Bert model, we conducted a careful study of two models: the Bert-base-Chinese model based on Chinese characters and the word-base-Chinese model based on Chinese words [36]. We found that the kernel used in the underlying word-base-Chinese model for Chinese words is "jieba," while the agri-food safety regulatory corpus, due to its domain specificity, was found to contain around 15% of subword errors when jieba subword tests were conducted. These errors have irreversible consequences for subsequent work. In contrast, the Bert-base-Chinese model based on Chinese characters does not require word separation, thus avoiding errors caused by boundary cuts and the sparsity problems associated with word separation. Moreover, it is essentially free of unregistered words and less prone to overfitting. Therefore, we continued to adopt the Bert-base-Chinese pre-trained language model for domain fine-tuning.

Fig. 4 EB-SJRE model architecture



3.2.2 Potential relationship forecast

The potential relationship prediction task in the field of agri-food safety regulation is primarily used to extract the types of potential relationships contained in the corpus. To accomplish this task, sequence context information is necessary. Furthermore, considering the length of the entities in the subsequent entity pair extraction task and the inversion of the subject and object positions, we integrated an RCNN layer before the relationship prediction. This layer comprises an LSTM model in both the forward and backward directions of the sequence and a maximum pooling layer to preserve important features of the text semantics while maintaining the sequence contextual context. To begin with, the relation prediction task maintains a corpus relation type embedding vector $E_r = (h_{r0}^d, h_{r1}^d, \dots, h_{rm-1}^d, h_{rm}^d)$, where the embedding vector h_{rm}^d corresponds to each relation in the corpus, and d is the embedding dimension, with rm denoting the relation type. This component is shown in the orange box in Fig. 4. For a given corpus S , the relationship judgment vector $R_s = (r_0, r_1, \dots, r_{m-1}, r_m | r_m \in (0,1))$ is obtained by the potential relationship prediction module, where m is the size of the set of relationship types. This component is shown in the red box in Fig. 4.

In contrast to previous work on repetitive entity extraction for each relation [22, 31], we first predict the subset of potential relations that may exist in the sentence and then perform the corresponding entity pair extraction under these subsets of potential relations. Given a sentence with embedding $h \in E^{n \times d}$, the formal definition is as follows:

$$h^{mid} = \text{Maxpool}(\text{Concat}(h, \text{BiLSTM}(h))) \quad (1)$$

$$R_s = \sigma(W_r h^{mid} + b_r) > \lambda : 1, 0 \quad (2)$$

$$R_k^{pot} = R_s E_r, k < m \quad (3)$$

where Concat is a join operation, BiLSTM is a bi-directional long and short-term memory network encoder to capture sequence semantic information, Maxpool is a maximum pooling operation to capture important features in the

S-embedding of the corpus. W_r is a trainable weight matrix of size $d \times m$, σ denotes the activation function, and b_r is the bias vector, this formal definition can be considered as a multi-label binary classification task, with each relation r_m in R_s set to 1 if its r_m probability exceeds some threshold λ , and 0 otherwise. R_k^{pot} is the final potential relation feature extracted, and k is the number of relation features obtained.

3.2.3 Entity pair extraction

The entity pair extraction task is performed only under the potential relationship condition, thereby avoiding the adverse consequences of dimensional explosion and label sparsity associated with relational entity modeling. For a given corpus S , after obtaining its potential relationship feature set R_k^{pot} , a relationship-specific entity pair labeling matrix T_{re} is constructed for each potential relationship to decode and determine the subject-object entity pair conditional on that relationship. However, the labeling scheme proposed in Sect. 3.1 only models the boundary association of the subject-object pairs of the triad at an intuitive level and does not have a deep semantic relationship. As shown in Fig. 5, "恩" is linked to "皮" and "呼" with an intuitive "H"; however, there is no deep semantic relationship between "恩" and "皮" and "呼". The discussion seems to have overlooked "皮" and "呼". Here, "恩" serves as the start marker of "恩诺沙星" (Enrofloxacin), "皮" is the start marker of "皮肤感染" (skin infections), and "呼" is the start marker of "呼吸道感染" (respiratory tract infections). Therefore, we propose the token pair boundary attention mechanism (TPA), which fuses potential relations and enables subject-object boundaries to be linked at the physical level while being reinforced by semantic boundary associations. The specific formal definition is as follows:

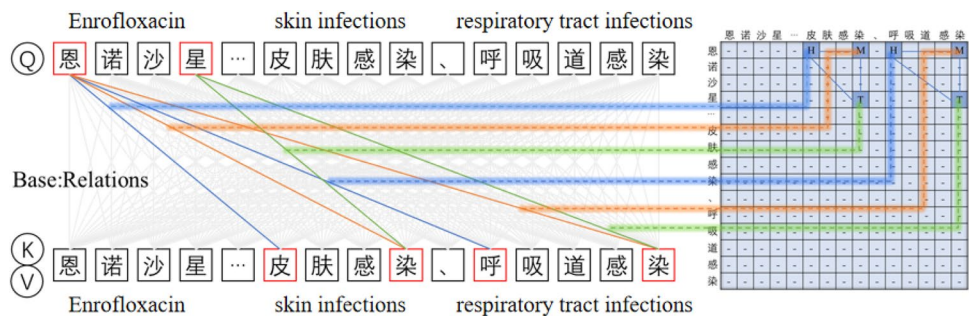
$$H_h = H_t = h_h + R^{pot} \quad (4)$$

$$\text{TPA}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_n)W \quad (5)$$

$$\text{head} = \text{Attention}(H_h, H_t, H_t) = \text{Softmax}(H_h H_t^T) H_t \quad (6)$$

where h_h denotes the vector corresponding to the sequence of corpus characters, H_h and H_t correspond to the head and

Fig. 5 TPA semantic boundary association attention mechanism



tail vectors incorporating the relational features, *Concat* is the merge operation, *head* is the attention head, W is the multi-head attention weight, Q is the query vector corresponding to H_h , K and V are the semantic vectors before and after the sequence corresponding to H_i . By following the approach described above, we can model the association between entity pairs and effectively establish semantic associations between entity boundaries, leveraging the extensibility of entity pair labels. After the completion of entity pair modeling, a series of activation functions will map the labels to the four labels proposed in Sect. 3.1, as formally defined below.

$$H_h = H_i = TPA(H_h, H_i, H_i) \quad (7)$$

$$T_{re} = re^T(H_h \odot H_i) \quad (8)$$

$$h_e = Relu(W_{re}T_{re} + b_{re}) \quad (9)$$

$$P(Tag_{i,j}|R^{pot}) = Softmax(W_{re}Drop(h_e) + b_{re}) \quad (10)$$

where $\odot$ denotes circular correlation, used to link potential dependencies between two character, re^T is the transpose function corresponding to the multi-headed token matrix, T_{re} is the multi-headed token matrix obtained by association, of size $[n \times n \times 2d]$. W_{re} is a trainable parameter matrix of size $[2d \times d]$, b_{re} is the corresponding bias, *Relu* is the activation function, h_e is the transition output, which has a size of $[n \times n \times d]$. W_{re} is a trainable parameter matrix of size $[d \times 4]$, b_{re} is the corresponding bias, *Drop* is used to speed up convergence and avoid overfitting, the corresponding activation function of this layer is *Softmax*, and the output $P(Tag_{i,j}|R^{pot})$ is the obtained tag probability matrix of size $[n \times n \times 4]$.

During the prediction stage, the label with the highest probability from $P(Tag_{i,j}|R^{pot})$ is extracted. Subsequently, the matrix of entity labels corresponding to potential relationships is generated, as illustrated in Fig. 3. Following this, the subject of the triad can be extracted based on M and T labels, and the object of the triad can be extracted based on H and M labels. Finally, the complete triad can be generated.

3.2.4 Training strategy

The joint training optimally combines the objective functions and shares the embedding layer encoder parameters. It

is important to note that the two tasks of potential relationship prediction and entity pair extraction under relationship correspondence are performed simultaneously. This is because Gold relationship labels are used when performing entity pair extraction, and therefore, the two tasks do not affect each other. As a result, the total loss can be divided into two parts, as follows:

$$L_{rel} = -\frac{1}{m} \sum_{i=0}^m (y_i \log R_s + (1 - y_i) \log(1 - R_s)) \quad (11)$$

$$L_{tag} = -\frac{1}{n^2 \times k_r^{pot}} \sum_{k=1}^{k_r^{pot}} \sum_{i=1}^n \sum_{j=1}^n y_{i,j} \log P(Tag_{i,j}|R^{pot}) \quad (12)$$

where m is the size of the corpus relation set, n is the length of the corpus sequence, k_r^{pot} is the size of the set of potential relations for the utterance, and the total loss is the sum of these two components.

$$L_{total} = L_{rel} + L_{tag} \quad (13)$$

4 Experiments

4.1 Dataset description

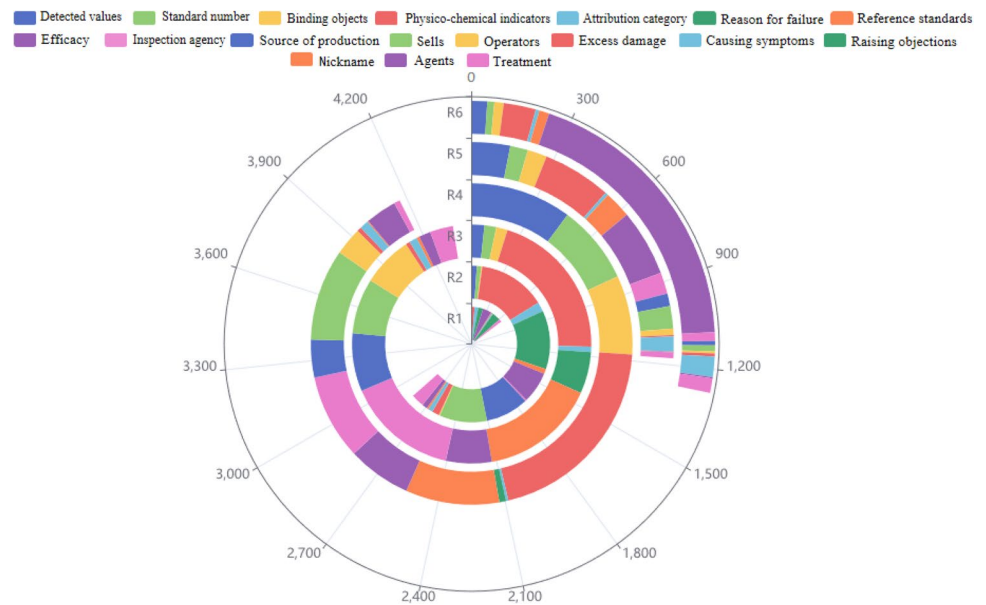
In this paper, relevant data was crawled from the websites of the Department of Agri-food Safety Standards and Inspection and Assessment of China and the Department of Agri-food Safety Sampling and Testing of the State Administration of Market Supervision and Administration. After cleaning and de-weighting, a total of 4877 samples were obtained, containing 18 relationship patterns and 14,207 triads. The relationship between the number of samples and the number of triads is illustrated in Table 2.

Figure 6 illustrates the distribution of the 18 relationship types in the samples with different numbers of included relationships. From the figure, it can be observed that the distribution of each relationship type is more even in the samples with 2, 3, and 4 included relationships. Conversely, the distribution of each relationship type in the samples with 1, 5, and 6 included relationships is variable and the total number of relationships is small. Therefore, both the number of relationships included in the samples and the distribution of the relationship types pose a challenge to our work.

Table 2 Agri-food safety regulatory corpus label correspondence table

Relation num	1	2	3	4	5	≥ 6	Total
Sample num	619	1385	1432	1035	231	159	4877
Triple num	619	2770	4296	4140	1155	1227	14,207

Fig. 6 Map of agri-food safety regulatory data relationship patterns



4.2 Evaluation metrics

In our experiments, we measure performance using three standard metrics: precision (P), recall (R), and the F1 score. To clarify, we define true positives (T_p) as the number of predicted triples that are correct, false positives (F_p) as the number of predicted triples that are incorrectly identified as existent, and false negatives (F_N) as the number of actual triples that the model failed to predict. Precision (P) is the percentage of true positives (T_p) among all positive results ($T_p + F_p$), indicating the model's accuracy for positive identifications. Recall (R) calculates the percentage of true positives (T_p) out of all actual positives ($T_p + F_N$), reflecting the model's capability to detect all relevant instances. The F1 score combines precision and recall into a single metric, representing the harmonic mean of the two, and is particularly useful for evaluating models when false positives and negatives are equally important. These metrics are calculated using the following criteria.

$$P = \frac{T_p}{T_p + F_p} \times 100\% \quad (14)$$

$$R = \frac{T_p}{T_p + F_N} \times 100\% \quad (15)$$

$$F1 = \frac{2PR}{P + R} \times 100\% \quad (16)$$

4.3 Model parameter setting

The effectiveness of the model is influenced by various parameters, such as learning rate, batch size, dropout rate,

number of model iterations, and more. A higher learning rate may cause the loss to fail to converge, while a lower learning rate may result in slower convergence or gradient disappearance. A larger batch size tends to make the model converge at the local optimum point, while a smaller batch size may cause the model to jump out of the saddle point. A suitable dropout rate can effectively alleviate overfitting and achieve regularization to a certain extent. Increasing the number of iterations generally improves the model's performance, but an excessive number of iterations can lead to a waste of time and overfitting, resulting in decreased model effectiveness. This experiment determined the most suitable parameter values using the control variable method, as shown in Fig. 7.

Figure 7a shows that the F1 score of the model continues to improve when the downstream learning rate is set to 5×10^{-3} , but too small of a learning rate may lead to lengthy training time and gradient loss, thus a moderate value of 5×10^{-3} is chosen as the learning rate of the model. Figure 7b shows that the F1 score is highest when the batch size is 12, which matches the text sequence size. Figure 7c shows that the model fits best when the dropout rate is set to 0.2. Figure 7d shows that the model's F1 score peaks at around 100 iterations. Based on the above findings, we finalize the parameter values as shown in Table 3.

It is essential to note that when fine-tuning the upstream pre-trained language model BERT on the domain dataset, the learning rate for fine-tuning differs significantly from the learning rate for downstream entity relationship decoding. The former requires a lower learning rate, while the latter requires a larger learning rate. Therefore, we implemented the model to separate the upstream and downstream tasks so that BERT can directly impact the downstream potential relationship recognition and entity decoding tasks, allowing

Fig. 7 The effect of model parameters on model performances

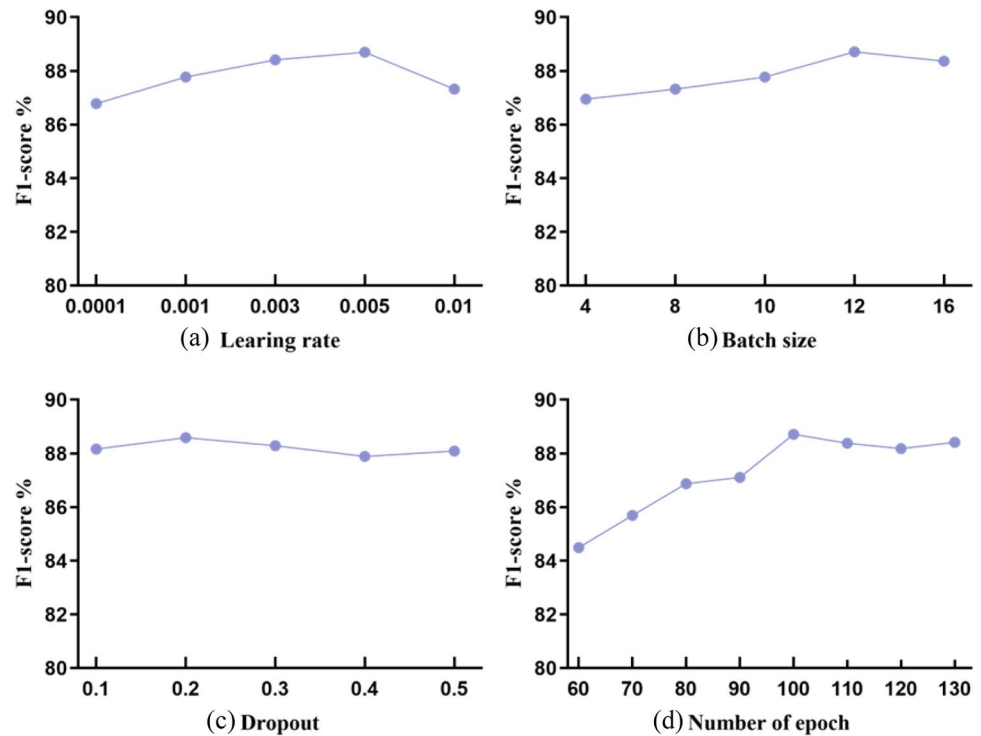


Table 3 Model parameter settings

Parameter	Value
Fine-tuning Bert learning rate	10^{-5}
Downstream task learning rate	5×10^{-3}
Warmup proportion	0.1
Number of epoch	100
Batch size	12
Weight decay rate	10^{-2}
Dropout rate	0.2
Word embedding dimension	768
Relation embedding dimension	768

Table 4 Results for all models on the agri-food regulatory domain dataset

Model	Agri-food safety regulatory field data		
	P (%)	R (%)	F1 (%)
CopyRE	68.50	50.45	58.10
MHSA	84.28	70.59	76.83
CasRel	88.54	85.37	86.93
PRGC	88.10	85.64	86.85
RIFRE	86.72	84.28	85.48
TP-Linker	87.28	83.39	85.29
OneRel	87.15	86.79	86.97
EB-SJRE	89.84	87.61	88.71

the downstream tasks to better influence the feedback to the BERT fine-tuning task.

4.4 Experimental results

4.4.1 Experimental results and analysis of field data

In this section, several typical and recently published entity-relationship federated extraction models are considered as comparison models, including CopyRE [37], MHSA [38], CasRel [22], PRGC [39], RIFRE [40], TP_Linker [31], and OneRel [30]. The models proposed in this paper were executed using the experimental parameter configurations mentioned above, while keeping the other seven model

parameters consistent with the original text. The specific extraction results are shown in Table 4.

The results indicate that EB-SJRE achieves a significant improvement in overall performance on the agri-food safety regulation domain dataset, with a qualitative improvement compared to traditional joint extraction schemes such as CopyRE and MHSA, resulting in an increase in F1 values of 28.04 and 11.88 percentage points, respectively. In the semi-joint extraction scheme, EB-SJRE outperforms CasRel, PRGC, and RIFRE by 1.78, 1.86, and 3.23 percentage points, respectively. In the end-to-end joint extraction scheme, EB-SJRE achieves 3.42 and 1.74 percentage point improvements compared to TP-Linker and OneRel, respectively. The reasons for these results are as follows:

1. Regarding the traditional joint extraction schemes CopyRE and MHSA, CopyRE typically combines the copying mechanism with the sequence structure to extract relations in the corpus first, and then identifies the current word with the highest probability as the entity under the relation. This scheme assumes a uniform prediction distribution when extracting entities, and there is no essential difference between the subject and the object. Therefore, the subject needs to be masked when extracting the object to prevent them from being consistent. Additionally, this scheme heavily relies on the effect of participles, as it can only target a single word. However, the domain corpus in this paper is extremely complex, and there are still a large number of combined entities after participles, which may contribute to its poor performance. On the other hand, MHSA comprises an entity extraction module and a relationship detection module. The entity extraction module uses CRF as the entity decoder to identify the subject and object in the sequence, while the relationship detection module uses a self-supervised attention mechanism to learn fine-grained semantic associations for each relation as a separate subspace. However, it ultimately combines the predictions of the two parts by thresholding to determine the triad, which may lead to greater uncertainty. To ensure accuracy, the threshold is limited; however, this also results in reduced recall.
2. The semi-joint extraction schemes of the day, such as CasRel, RIFRE, and PRGC, are popular extraction patterns that establish some kind of association. CasRel and RIFRE both use a cascaded binary tokenization framework to model relations and objects in two dimensions. Typically, they first identify the subject and then incorporate the subject features to identify the corresponding objects under specific relationship conditions. The difference between the two is that RIFRE introduces a heterogeneous graph attention mechanism to model the association of tokens and relations in the corpus. However, on this dataset, RIFRE does not seem to work well and even has slight performance degradation compared to the simple and brute-force CasRel. We speculate that this may be due to the presence of more entities in this dataset, and a large number of entity-relationship association modeling may lead to distortion of the sequence semantic information, resulting in performance degradation. PRGC, on the other hand, is more like a pure pipeline extraction model. It first extracts the relations in the corpus, maps the Token Pair under the relations, and then extracts the subject and object entities by traditional BIO annotation. Finally, it determines the triad by mapping between the entities and the Token Pair. However, this approach has more steps and exposes more bias. Moreover, the BIO annotation approach is not suitable for dealing with a large number of various entities such as "0.58 mg/kg" in the corpus.
3. Regarding the "real" federated extraction solutions, TP-Linker and OneRel can avoid bias exposure caused by multi-segment extraction and achieve end-to-end entity relationship extraction. Both of them are spatial models of the triads in the corpus, directly identifying the subject-object association information under the corresponding relational features. They then decode the subject-object association information to determine the triads. While their general idea is the same, the difference lies in the design of the labels. OneRel is more intuitive in aligning the beginning and end of the two entities at the head and end, while TP-Linker extends the link between entities on a single entity-wide label. From a machine perspective, they are identical. However, their results are mediocre on this dataset. We analyzed that this may be due to the fact that the data in this paper has up to 18 types of relationships. Direct spatial modeling of relationships and entities would cause problems such as sparse labels, which would affect model accuracy and recall.
4. CasRel's semi-joint extraction structure and OneRel's idea of entity association modeling serve as inspiration for EB-SJRE, and its overall effect is not significantly different from these two models in this regulatory domain data. However, the combined tagging and overall performance metrics show that CasRel has lower recall compared to OneRel and EB-SJRE, resulting in a lower overall F1 metric than both. This indirectly indicates that the cascaded binary annotation scheme relying on CasRel does not perform as well in practice as models such as OneRel and EB-SJRE, which use Token pair correlation modeling. Combining the model structure and overall performance, EB-SJRE shows a small improvement in both accuracy and recall compared to CasRel and OneRel, both of which model relationships and entities spatially, thereby building some kind of correspondence link. However, this approach also brings the problem of increased training parameters and sparse labels. As a semi-joint extraction scheme, EB-SJRE has a small exposure to bias due to relational prediction, but it can be further optimized through parameter control. In contrast, label sparsity and increased parameters are problems at the model architecture level that are largely insurmountable. From this perspective, EB-SJRE is also advantageous at the level of model label architecture.

4.4.2 Ablation experiments

To verify the effectiveness of the RCNN [24] encoder and the TPA attention mechanism, we selected RNN, LSTM, Bi-LSTM [41], GRU, Bi-GRU [42], and 2D-CNN [43] feature

extractors for comparative ablation experiments. The results are shown in Table 5. The Base model has not added any encoder after Bert encoding, and all models were replaced using fully connected layers.

The results demonstrate that adding RNN, GRU, and LSTM encoders in the relational prediction stage can improve the model's performance to some extent. The main index of improvement is the recall rate, which increases by 2.22, 1.71, and 1.88 percentage points, respectively, compared with the Base model. However, the overall accuracy rate drops slightly, which we attribute to the fact that the one-way recurrent neural class network fails to effectively cover the semantic features of the domain sequence Token during sequence feature extraction. This affects the accuracy rate of entity pair extraction and leads to a lower global accuracy rate. In contrast, overall performance improves significantly when replaced with Bi-GRU and Bi-LSTM. Although the overall accuracy rate decreased slightly compared to the Base model, the recall rate improved by 2.61 and 3.68 percentage points, respectively. Replacing the encoder with 2D-CNN resulted in a slight improvement in recall and a slight decrease in accuracy, with a slight improvement in overall F1. Replacing the RCNN encoder improved recall by 2.9 percentage points, precision by 0.58 percentage points, and F1 by 1.79 percentage points compared to the Base model. This improvement was achieved because the RCNN kernel consists of a Bi-LSTM and a CNN. The Bi-LSTM can obtain as much contextual semantics of the supervised corpus as possible, while the CNN maximum pooling works to better determine which features in the text can play a more important role in entity-to-boundary recognition.

After adding the TPA boundary attention module in the entity prediction stage, the accuracy improves by 0.32 percentage points, the recall improves by 0.95 percentage points, and the overall F1 value improves by 0.64 percentage points compared to adding only RCNN. This indicates that the sequence information incorporated with relational features can effectively capture the entity boundary association

features under relational conditions after the TPA self-attention mechanism. As a result, the overall performance indexes are all improved to a certain extent. These results validate that incorporating RCNN and TPA mechanisms in EB-SJRE can effectively improve the overall effect of relationship extraction.

4.4.3 Detailed experiments

Table 6 displays the extraction effects of EB-SJRE for different relationships in the regulatory domain data. Since CasRel, OneRel, and EB-SJRE only differ by approximately 2 percentage points in the overall extraction results shown in Table 4, and the semi-joint, entity-pair association properties of EB-SJRE are also inspired by both of them, this experiment includes them in the specific extraction effect comparison.

The results showed that EB-SJRE outperformed CasRel in 10 groups of relationships, including 理化指标 (Physico-chemical indicators), 检测值 (Detected values), 归属品类 (Attribution category), 不合格原因 (Reason for failure), 检验机 (Inspection agency), 功效 (Efficacy), 生产源头 (Source of production), 运营商 (Operators), 引起症状 (Causing symptoms), and 治疗 (Treatment). Additionally, EB-SJRE outperformed OneRel in 10 groups of relationships, including 约束对象 (Binding objects), 不合格原因 (Reason for failure), 参考标准 (Reference standards), 检验机构 (Inspection agency), 功效 (Efficacy), 生产源头 (Source of production), 运营商 (Operators), 过量损害 (Excess damage), 引起症状 (Causing symptoms), and 治疗 (Treatment). In the three groups of relations, 不合格原因 (Reason for failure), 功效 (Efficacy), and 引起症状 (Causing symptoms), EB-SJRE has 1.42, 8.57, and 2.81 percentage point improvements in F1 values compared to OneRel, and 8.38, 4.44, and 5.92 percentage point improvements in F1 values compared to CasRel. Most of the entities of these three relationships exist in the form of one-to-many or many-to-many in the corpus, which directly indicates that

Table 5 Impact of different encoders on model composite metrics

Model		Agri-food safety regulatory field data		
		P (%)	R (%)	F1 (%)
Potential relationship forecast	Base	88.94	83.76	86.28
	+RNN	87.19	85.98	86.58
	+GRU	88.73	85.47	87.07
	+LSTM	88.10	85.64	86.85
	+Bi-GRU	88.56	86.37	87.45
	+Bi-LSTM	88.28	87.44	87.86
	+2D-CNN	87.48	85.16	86.31
	+RCNN	89.52	86.66	88.07
Entity extraction	Base + RCNN	89.52	86.66	88.07
	Base + RCNN + TPA	89.84	87.61	88.71

Table 6 Agri-food safety regulatory data entity relationship identification results

Relationship Model		EB-SJRE			CasRel			OneRel		
		P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)
理化指标	Physico-chemical indicators	90.38	89.76	90.07	88.28	88.89	88.58	89.45	91.32	90.38
检测值	Detected values	89.23	79.45	84.06	87.41	80.82	83.99	85.31	83.56	84.43
约束对象	Binding objects	91.75	87.25	89.45	94.68	87.25	90.82	89.00	87.25	88.12
标准编码	Standard number	94.23	97.03	95.61	96.12	98.02	97.06	96.11	98.02	97.06
归属品类	Attribution category	72.09	73.81	72.94	75.00	64.29	69.23	75.00	78.57	76.74
不合格原因	Reason for failure	85.71	82.11	83.87	81.21	70.53	75.49	83.33	81.58	82.45
参考标准	Reference standards	94.70	94.04	94.37	95.05	94.39	94.72	92.83	95.43	94.12
检验机构	Inspection agency	97.71	97.71	97.71	97.70	97.33	97.51	97.70	97.70	97.70
功效	Efficacy	79.20	73.30	76.14	72.75	70.68	71.70	68.00	66.96	67.47
生产源头	Source of production	91.63	93.00	92.31	89.32	92.00	90.64	89.65	91.00	90.32
销售	Sells	96.75	95.37	96.06	96.42	95.73	96.07	96.76	95.73	96.24
经营商	Operators	94.90	95.08	94.99	93.88	94.85	94.36	93.87	94.84	94.35
过量损害	Excess damage	73.68	70.00	71.79	86.67	65.00	74.29	68.42	65.00	66.67
引起症状	Causing symptoms	86.67	82.54	84.55	85.19	73.02	78.63	90.38	74.60	81.74
提出异议	Raising objections	97.44	95.00	96.20	97.44	95.00	96.20	97.43	95.00	96.20
别称	Nickname	100.00	66.67	80.00	85.71	80.00	82.75	100.00	66.67	80.00
代理商	Agents	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
治疗	Treatment	80.00	66.67	72.73	77.36	68.33	72.56	72.54	61.66	66.66

EB-SJRE has better performance than CasRel and OneRel when dealing with complex situations such as entity overlap. Additionally, EB-SJRE only has F1 values below 80% for three types of relationships: 归属品类 (Attribution category), 过量损害 (Excess damage), and 治疗 (Treatment). Half of the relationships have F1 values above 90%, which makes it more applicable to the regulatory data domain constructed in this paper.

Furthermore, since the data in the agri-food safety regulation domain contains complex information, a corpus sentence will typically contain more than one helpful piece of information. Therefore, the model must have strong multi-relational entity processing capability. The experiment classified the corpus into six categories based on the number of triads and introduced CasRel and OneRel models as a comparison. The results are shown in Table 7.

The results showed that EB-SJRE outperformed CasRel and OneRel in all six categories of the corpus, especially in

the corpus containing six or more relationships. In this category, EB-SJRE showed an increase of 5.01 and 10.09 percentage points in F1 value compared to CasRel and OneRel, respectively. These results demonstrate that EB-SJRE can effectively handle agri-food safety regulatory data containing multiple relationships without being overwhelmed by a corpus containing multiple pairs of relationships.

4.4.4 Generalisation experiments

To verify the generalization ability of the EB-SJRE model, we conducted experiments on four public datasets: NYT (New York Times dataset) [44], NYT-Star, WebNLG (Web Natural Language Generation dataset) [45], and WebNLG-Star. NYT and WebNLG are two widely used public datasets. We also use their variants, NYT-star and WebNLG-star, in our study. They represent specific annotation methods for these datasets, respectively. We also

Table 7 Extraction results of the model on a corpus containing different numbers of relations

Number of relations	%	EB-SJRE			CasRel			OneRel		
		P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)
1	13.89	73.79	75.89	74.83	72.93	68.79	70.80	71.71	77.30	74.40
2	26.40	88.16	84.70	86.39	86.02	81.53	83.72	85.58	84.14	84.85
3	29.75	91.56	91.06	91.31	91.54	89.62	90.57	89.10	88.47	89.74
4	21.87	94.33	91.78	93.04	92.90	91.33	92.11	93.21	92.79	93.00
5	4.93	89.75	87.60	88.66	82.76	86.40	84.54	87.06	88.80	87.92
≥ 6	3.06	79.38	70.97	74.94	74.48	65.90	69.93	70.05	60.37	64.85

included several typical entity-relationship joint extraction models that have been proposed in recent years, such as CopyRE, MHSA, TP_Linker, RIFRE, PRGC, CasRel, and OneRel. All experimental data were based on Bert-base-Chinese, and the specific environment configuration was 8 cores and 32G NVIDIA V100 GPU, PyTorch 1.8, and CUDA 10.1. The results of these experiments are shown in Table 8.

The results indicate that under the specified environment configuration, EB-SJRE achieved the best overall F1 metrics on NYT and NYT-star, with a qualitative improvement over traditional CopyRE and MHSA, especially on NYT-star. In this dataset, EB-SJRE showed a 3.2 percentage point improvement compared to the typical CasRel in recent years. Compared to other joint extraction models published in recent years, although they did not make significant progress, the results are still in the first-tier echelon. On WebNLG, EB-SJRE exhibited a qualitative improvement over CopyRE and a slight lead over other joint extraction models published in recent years. On WebNLG-star, EB-SJRE ranked second in terms of overall F1 metrics, only behind OneRel. However, it is comparable to all the joint extraction models published in recent years. Therefore, if training speed is a priority, EB-SJRE and PRGC are recommended based on their semi-joint structure. For those seeking the ultimate performance metrics, EB-SJRE and OneRel should be considered, but the implementation of OneRel in specific scenarios should be examined in conjunction with the length of the data corpus and the number of relationships, as it may occupy a vast amount of memory space.

Additionally, we found that EB-SJRE demonstrated better accuracy rates on the NYT and WebNLG corpus, while the recall rates were comparable to other models. Moreover, it can reduce a considerable part of the error correction workload if it needs to be applied to downstream tasks. These experiments demonstrate that EB-SJRE has excellent generalization capabilities and can be applied not only to entity relationship extraction in the agri-food safety regulation domain but also in other domains.

5 Conclusion

To address the challenges of relationship variety, complex entity composition, and overlapping entity relationships in entity-relationship extraction for agri-food safety regulation, this thesis proposes a semi-associative entity relationship extraction model based on entity pair boundary delineation. Firstly, a set of Token-pair-level subject-object entity association labels is designed for entity pair boundary identification, enabling direct modeling of the triadic subjects and objects present in the corpus. Effectively mitigates the challenges posed by the diversity and complexity of entity relationships. Secondly, the architecture of the semi-associative relationship extraction model is utilized to improve the inference speed of relationship extraction. It reduces bias exposure and relieves problems such as dimensionality expansion that can occur in joint extraction. Furthermore, the entity pair boundary attention mechanism, TPA, is discussed to effectively capture the semantic association features of the entity boundaries of the text segments. Extensive experiments were conducted on the agri-food safety regulation domain dataset and four public datasets, NYT, NYT-star, WebNLG, and WebNLG-star, and the results demonstrate that the proposed model achieves significant improvements on the domain data. The ablation experiments justify the component encoders used, while the considerable advantages on the public datasets also prove its ability to generalize well.

However, there are still shortcomings in the current work. In future work, we plan to explore more features that are more useful for entity relationship extraction, such as syntactic analysis. We also hope to use this task to explore downstream application tasks of knowledge graphs. Additionally, the tagging architecture for joint entity-relationship extraction will be a research focus in the future. Complementing these developments and in line with the insights from Akkem et al. [46], we will also prioritize the continuous monitoring of our model. This will involve utilizing MLOps techniques for ongoing performance

Table 8 Extraction effects of various models on different public datasets

Model	NYT			NYT-Star			WebNLG			WebNLG-Star		
	<i>P (%)</i>	<i>R (%)</i>	<i>F1 (%)</i>	<i>P (%)</i>	<i>R (%)</i>	<i>F1 (%)</i>	<i>P (%)</i>	<i>R (%)</i>	<i>F1 (%)</i>	<i>P (%)</i>	<i>R (%)</i>	<i>F1 (%)</i>
CopyRE	61.04	57.10	59.00	—	—	—	38.50	36.32	37.38	—	—	—
MHSA	88.08	78.35	82.93	—	—	—	89.85	86.10	87.93	—	—	—
TP_Linker	92.82	91.18	92.20	91.18	92.39	91.78	88.90	84.50	86.70	91.80	92.00	91.90
RIFRE	93.12	90.16	91.62	93.60	90.50	92.00	—	—	—	—	—	—
PRGC	92.65	91.9	92.27	92.36	92.01	92.21	89.90	87.20	88.50	91.62	91.27	91.44
CasREL	—	—	—	89.70	89.50	89.60	—	—	—	93.40	90.10	91.80
Onerel	91.99	92.27	92.13	93.10	92.07	92.58	90.17	87.55	88.84	92.40	93.80	93.09
EB-SJRE	93.27	91.47	92.36	92.80	92.90	92.80	91.22	86.70	88.91	91.63	92.79	92.21

evaluation and adaptation, ensuring our model remains accurate and relevant in changing environments.

Acknowledgements This work was supported by the Beijing Science Foundation (9232005); the Social Science and Humanity on Fund of the ministry of Education (23YJAZH216).

Data availability statement The initial data for this study was obtained from the websites of the Agricultural Product Food Safety Standards and Inspection Testing Evaluation Department and the Agricultural Product Food Safety Sampling and Inspection Department of the State Administration for Market Regulation in China. The data underwent cleaning, removal of duplicate entries, and manual annotation. However, due to data licensing restrictions, we are unable to publicly share the annotated data. If there are reasonable requests, we will provide the data to the relevant authors.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Consent for publication Not applicable.

Ethics approval and consent to participate Not applicable.

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